2023 9th Annual International Conference on Network and Information Systems for Computers (ICNISC) | 979-8-3503-0668-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/ICNISC60562.2023.00085

Sensor Optimization in Large Spaces Based on RBF Neural Networks and K-means Cluster Analysis

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***Abstract*—Large-scale spatial buildings are widely prevalent across the nation; however, due to their size, accurate temperature measurement is challenging, resulting in increased energy consumption for air conditioning. Based on this, this paper takes the check-in hall of a certain airport terminal in Shanghai as the research subject. It analyzes temperature key points in the check-in hall partitions and employs a numerical simulation approach (CFD) using the Airpark software for temperature cloud mapping analysis. Furthermore, it utilizes the RBF neural network for sensor quantity prediction analysis. Finally, the k-means clustering analysis method is applied to determine the sensor placement locations. The results indicate that after optimization, the economic cost of sensors has decreased by 51.55%, and the average absolute error has also reduced.**

Keywords—RBF neural network, k-means clustering analysis, style, large-span spatial structures, optimal sensor placement

I. INTRODUCTION

Balancing the demands between comfort and energy consumption is a challenging and focal point in energy conservation for large spaces. Among these, precise and accurate temperature feedback is one of the key factors in resolving this issue. Xu Songqi [2] utilized the Discrete Firefly Algorithm for sensor optimization in docks, concluding that this method is more effective than the traditional BPSO method. This paper draws inspiration from their discrete approach and makes improvements.Xiu Xiaobo [3] used a GA algorithm for sensor placement in a fan drying environment, optimizing it with the RBF algorithm. A comparison was made to calculate the optimal cost-effectiveness of sensors, reducing the average absolute error and economic cost during measurement.Xia Jing [4] conducted an analysis and monitoring of vulnerable points in bridges, subsequently performing cluster analysis to determine suitable sensor locations and their effective independence. Damage-related information was collected based on parameters.Lv Yingdi [5] used the Monkey Swarm Algorithm to validate and optimize the placement and quantity of single sensors. Following this, a neural network model was constructed for modeling and calculations.

Combining the aforementioned algorithms and optimization methods, this paper takes the example of an airport check-in hall and conducts an analysis and research on airport cooling supply. The analysis includes simulation and optimization using CFD technology. Additionally, the coupling of the RBF neural network and k-means algorithm is applied to optimize the placement of temperature sensors at a certain Shanghai airport.

II. MATHEMATIC MODEL

A. k-means algorithm

Step 1: Determine the number of sensors, K.; Step 2: Randomly select an arbitrary point as the centroid of the system, and calculate the distance (Euclidean distance) between each temperature key point and the random centroid using the algorithm. Use this result to assign temperature key points to the same category.; Step 3: Iteratively calculate the new centroid for each cluster, and when the centroids no longer change, the process converges.

The given dataset (coordinates of temperature key points) contains i objects, with each object being a point in a two-dimensional plane.

$$X = \{X_1, X_2, X_3, X_4, \dots\dots, X_i\} \quad (1)$$

Assign the objects from the dataset to randomly generated cluster centers, which include K objects. Each object is a point within a two-dimensional plane.

$$C = \{C_1, C_2, C_3, C_4, \dots\dots, C_k\} \qquad (0 < k < i) \quad (2)$$

When assigning key points to different clusters, the nearest-neighbor measurement strategy is commonly used. In Euclidean space, the distance utilized is the Euclidean distance. When dealing with document formats, cosine similarity or Manhattan distance is often employed. Euclidean distance:

$$d(x,c) = \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2 + \cdots + (x_n - c_n)^2} = \sqrt{\sum_{i=1}^{n} (x_i - c_i)^2} \quad (3)$$

cosine similarity: $\cos\theta = \frac{A \cdot B}{\|A\| * \|B\|}$ (4)

In the equation, A and B represent vectors (x_1, y_1) and (x_2, y_2) respectively. The numerator is the dot product of A and B, and the denominator is the square root of the sum of the squares of x and y.

Manhattan distance: $d_{12} = \sum_{k=1}^{n} |x_{1k} - x_{2k}|$ (5)

By sequentially comparing the distances of several cluster centers, they are allocated to K clusters, resulting in S = {S_1, S_2, S_3, S_4, …, S_k}. The definition of the cluster center is the mean of all elements in each dimension of the cluster, expressed as:

$$C_t = \frac{\sum x_i \in s_i X_i}{|S_i|} \quad (6)$$

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In the equation, C_l represents the center of the first cluster in the clustering, where $1 \leqslant l \leqslant k$.

B. RBF neural network

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Central Idea: Using RBF (Radial Basis Function) hidden units as the foundation, with the input data being coordinate points, it transforms low-dimensional data into high-dimensional data. This transformation allows for images that cannot be displayed in two-dimensional space to be linearly mapped into a high-dimensional form, where the output data is a linear output of the hidden units.

Width: The width in the function represents the effective range of the corresponding input data. The smaller the width, the smaller the effective range of the corresponding hidden neuron function, and the smaller the response of the neighboring neurons corresponding to that neuron. The expression is as follows:

$$d_{jt} = d_\rho \sqrt{\frac{1}{N}\sum_{K=1}^{N}(x_i^k - c_{jt})^2} \tag{7}$$

Center: The centers arrange the initial values in the hidden layer in equal intervals based on their magnitude, ensuring that even the weakest signal input can generate an ideal response here. The advantage of this approach is that different input values can quickly be found at their respective centers, highlighting the benefits of the Gaussian kernel. The expression is as follows:

$$c_{jt} = mini + \frac{maxi \quad mini}{2p} + (j-1)\frac{maxi \mid mini}{p} \tag{8}$$

Where j represents the index of the hidden layer neuron.; i represents the index of the input neuron.; mini represents the minimum value among the i input data.; maxi represents the maximum value among the i input data.

$$z_j = exp(-||\frac{X - C_j}{D_l}||^2) \tag{9}$$

Where Cj represents the center vector of the j-th data among i data points, which is composed of all neurons in the input layer corresponding to the j-th neuron function in the hidden layer. It can be expressed as: $C_j = [C_{j1}, C_{j2}, .., C_{jn}]$ Dj represents the width vector of the j-th data among i data points, which corresponds to C_j. A larger Cj implies a larger Dj, indicating better stability.

$$y_k = \sum_{i=1}^{p} w_{kj} z_j \tag{10}$$

k is the number of neuron functions in the output layer.

Weight Iteration:The algorithm employs gradient descent to train the function, adjusting and updating the center, width, and weights adaptively through learning.

$$w_{kj}(t) = w_{kj}(t-1) - \eta\frac{\partial E}{\partial w_{kj}(t-1)} + \alpha[w_{kj}(t-1) - w_{kj}(t-2)] \tag{11}$$

$$c_{kj}(t) - c_{kj}(t-1) - \eta\frac{\partial E}{\partial c_{kj}(t-1)} + \alpha[c_{kj}(t-1) - c_{kj}(t-2)] \tag{12}$$

$$d_{kj}(t) = d_{kj}(t-1) - \eta\frac{\partial E}{\partial d_{kj}(t-1)} + \alpha[d_{kj}(t-1) - d_{kj}(t-2)] \tag{13}$$

where η represents the learning rate.; E represents the mean squared error (loss function).

III. Prepare Your Paper Before Styling

In a three-dimensional space, measuring the distribution of temperature is challenging. Typically, it is segmented into multiple planes, and a few representative planes are selected for analysis. Given the characteristics of large-space buildings, it is known that people are most sensitive to temperature in the range of y=0 to 1.6 meters, and there is a significant vertical gradient in temperature in large-space buildings. Air conditioning outlets are generally located in the lower half, where temperature fluctuations are quickly reflected. Sensor detection radius is typically a few meters. Therefore, in this paper, we choose y=1.6 meters and y=0.8 meters as the research planes.

A. Sensor selection

For the airport check-in hall, there are many sensors to choose from. Here, we have selected the MD-HT100 series temperature sensors with a detection radius of 10 meters.

TABLE I. SENSOR SELECTION

Title	***model***	***measurement unit***	***measuring range***
temperature sensor	MD-HT100	℃	-40~+80℃
precision	***resolution ratio***	***response time***	***way to install***
±0.2℃	0.1℃	5s	one-piece

Reasons for the selection: The MD-HT100 sensor was chosen due to its high precision, durability, strong stability, excellent interference resistance, and isolated output signal. Its compact design allows for easy wall mounting, and it features functions such as unit switching and adjustable display digits.

B. Method

Using Air Park, a temperature contour map of the indoor environment in the airport check-in hall was simulated. Key positions were selected, including points with significant temperature fluctuations, the lowest and highest temperature points, points with substantial temperature variations, and areas where passengers tend to gather. An RBF neural network was trained on these key positions to determine the number of sensors. Subsequently, MATLAB 2019b was used for cluster analysis simulations.

This paper is based on the finite volume method, assuming that the air in the large space is three-dimensional,

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incompressible, steady-state, turbulent, and an ideal gas. Previous research has shown that the RNG k-ε model is more accurate than the standard k-ε model and is better suited for studying turbulence issues in indoor environments of large-space buildings [6]. Therefore, this paper selected the RNG k-ε model as the turbulence model.

physical model: This paper selects the departure hall of a certain airport terminal in Shanghai as the research subject. When simulating the construction, its overall shape is considered as a rectangular hall with dimensions of 414 meters in length and 140 meters in width. In the center, there are ten ticketing areas, each measuring 60x15 meters. The floor plan and three-dimensional view are shown below:

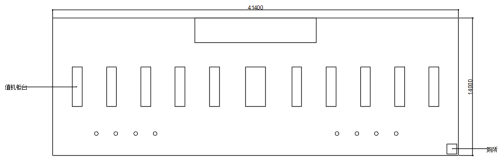


Fig. 1. Airport terminal plan.

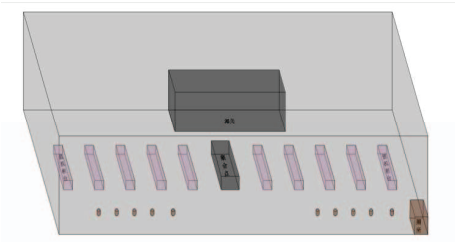


Fig. 2. Airport terminal three-dimensional diagram.

Boundary Conditions: For the research object, a quarter of the entire departure hall is selected, specifically an island for ticketing with dimensions of 207 meters in length and 70 meters in width. Heat transfer between wall surfaces in the large space is assumed to be uniform. Considering steady-state heat transfer, the second type of boundary conditions is chosen for heat transfer boundary conditions. Comprehensive radiation conditions are considered and calculated based on the comprehensive conditions.

TABLE II. BOUNDARY CONDITIONS

effective temperature difference(℃)	*Air supply speed (v/m)*	*Air supply Angle*	*turbulence model*
7	7.9	0	RNG k-ε model

C. Temperature cloud map

The temperature distribution cloud map of the cross section is as follows:

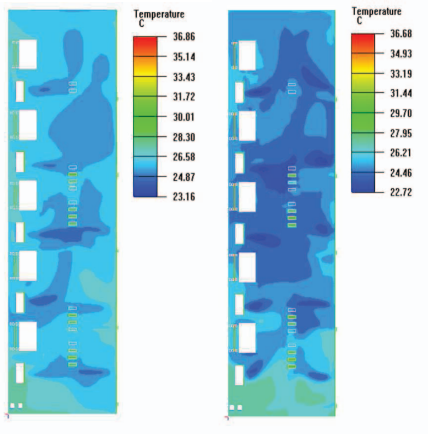


Fig. 3. Temperature cloud image at Y=1.6m(left) & at Y=0.8m

IV. CASE ANALYSIS

From the temperature distribution contour map plane at Y=1.6 meters, 80 key points were selected, and their temperature coordinates were used as initial data for calculations. After applying RBF neural network optimization calculations, the results are as follows:

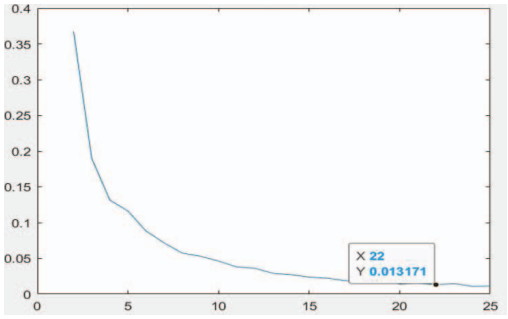
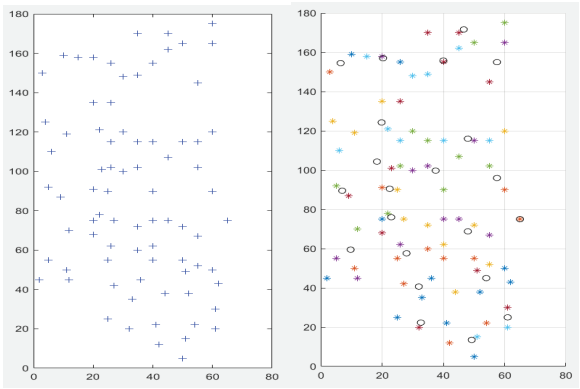


Fig. 4. RBF Neural Network Optimization at Y=1.6 meters

In the graph, the horizontal axis represents the number of clustering classes, and the vertical axis represents the average absolute error. It can be observed that although a greater number of sensors leads to a smaller average absolute error, considering cost-effectiveness and convenience, the best performance is achieved at x=22. In other words, when there are 22 sensors, the average absolute error is minimized, at 0.013171. This results in the generation of 22 cluster centers:



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Fig. 5. Temperature key point at Y=1.6m(left) & sensor location

At Y=0.8 meters:Using its coordinates as the initial data for calculations and after applying RBF neural network optimization, the results are as follows:

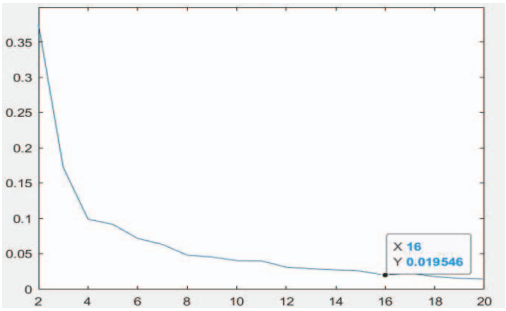


Fig. 6. RBF Neural Network Optimization at Y=0.8 meters

Similar to the case at Y=1.6 meters, at Y=0.8 meters, considering cost-effectiveness and convenience, the best performance is achieved at x=16. In other words, when there are 16 sensors, the average absolute error is minimized, at only 0.019546. Moreover, when sensors are evenly distributed, 33 sensors are required, but after optimization, only 16 sensors are needed. This represents a 51.55% reduction in sensor economic costs. Similarly, 16 cluster centers are generated, focusing on points with significant temperature fluctuations, and with a sensor detection radius of 10 meters.

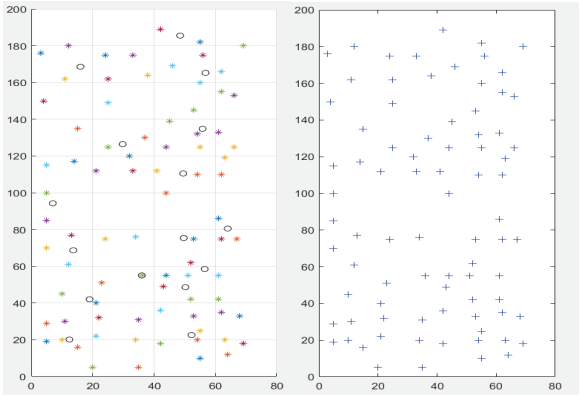


Fig. 7. Temperature key point at Y=0.8m(left) & sensor location

Different plane clustering summary:

TABLE III. DIFFERENT PLANE CLUSTERING SUMMARY

	Cluster center	*Sensor radius (m)*	*mean absolute error*
Y=1.6m	22	10	0.013171
Y=0.8m	16	10	0.019546

Analysis:

The entire process involves using Air Park to simulate the temperature contour map of a large space, selecting key points sensitive to temperature, estimating the number of sensors using the RBF neural network algorithm, and then optimizing the sensor placement. It can be observed that by introducing this comprehensive process and verifying it through algorithms and simulations, the improved method not only reduces the number of sensors but also maximizes the coverage of as many monitoring points as possible, thus maximizing the efficiency of sensor utilization.

V. CONCLUSION

This paper takes the check-in hall of a certain airport in Shanghai as an example and uses Air Park to conduct CFD simulations to generate temperature and humidity contour maps on different planes within the hall. Considering sensor coverage, timely and effective data feedback, and changes in human comfort, the following conclusions have been drawn:

(1) Through CFD simulations and the RBF and K-means methods, the optimal sensor placement positions have been determined, reducing the system's energy consumption while ensuring monitoring. When sensors are evenly distributed, 33 sensors are required. However, at Y=1.6 meters, after optimization, only 22 sensors are needed, resulting in a 33.33% reduction in economic cost, with an average absolute error of 0.013171. At Y=0.8 meters, after optimization, only 16 sensors are needed, reducing the economic cost by 51.55%, and the average absolute error is only 0.019546.

(2) The arrangement of the sensor quantity model forms the basis of the calculation. During subsequent feedback data analysis, different sensor quantities and positions can affect the stability and accuracy of the entire control system. Therefore, selecting the correct sensor quantity and position is crucial for optimization.

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